

MB 520: Fundamentals of Data and Analytics

Course Code	MB 520
Course Name	Fundamentals of Data and Analytics
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	MBA
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- **Data and Analytics Concepts:** Data concepts - DIKW and data analytics pyramid, small data to big data, Data analytic thinking, uncertainty and decision, data driven and goal driven decision making, Analytics processes and systems, data and analytics maturity, CRISP-DM Process. (4 Lectures)
- **Models and Processes:** Business analytics models, Strategy creation and Key Performance Indicators (KPIs), Business questions and KPIs, Asking right business questions on data and analytics, Data and Analytics Models and their types. (4 Lectures)
- **Data Preparation:** Data collection and preparation, perspectives on data, data types, sources and quality, data description using levels of measurement and types of measurement scales, Types of attributes/features, data cycle-the data preparation activities, data cleaning and data transformation, data encoding, data discretization, transformation for normality, feature selection- shrinking, dimensionality reduction. (6 Lectures)
- **Exploratory Analytics:** Describing the past, data visualization, understanding your data sources, understanding variability in the data. (4 Lectures)
- **Predictive Analytics:** Classes of predictive models-logic driven and data driven, predicting numerical and categorical values, asking predictive questions, simple and multiple linear regression as a predictive tool, correlation and multiple regression analysis, Forward and backward step-wise regression, Role of F-ratio and R-square adjusted statistics for predictive analytics (multiple regression based). (6 Lectures)
- **Prescriptive Analytics:** Optimization and experimentation for prescriptive analytics, asking prescriptive questions, optimization (MS Excel solver/other optimization tools), Prescriptive steps in analytics – defining the problem, decision variables, objective functions, constraints, and arriving at business solution. Introducing Causality, importance of causal analytics for business problem solving. (4 Lectures)

Text Books

NA

References

1. Daniel, Vaughan, **Analytical Skills for AI & Data Science**, Shroff Publishers and Distributors Pvt. Ltd, 2020.
2. Daniel T. Larose, Chantal D. Larose, **Data Mining and Predictive Analytics**, Wiley, 2016.
3. **HBR Guide to Data Analytics Basics for Managers**, Harvard Business Review Press, 2018
4. Provost, F and Fawcett, T., **Data Science for Business**, Shroff Publishers and Distributors Pvt. Ltd, 2014.
5. Jeffrey D. Camm, James J. Cochran, Michael J. Fry, Jeffrey W. Ohlmann, **Business Analytics: Descriptive, Prescriptive and Predictive**, 4th Edition, Cengage Learning Inc, 2021.
6. Laursen, G.H.N. and Thorlund, J., **Business Analytics for Managers**, Wiley India Pvt. Ltd., 2014.
7. Amar Sahay, **Essentials of Data Science and Analytics Statistical Tools, Machine Learning, and R-Statistical Software Overview**, Business Expert Press, 2021.
8. J. D. Kelleher and B. Tierney, **Data Science**, The MIT Press, 2018.